1 **(a)** $\dfrac{\sqrt{3}}{2}$ **[B1]**

(b) $\sin 30° = \dfrac{1}{2}$ and $\tan 60° = \sqrt{3}$ **[M1]**

$\dfrac{1 - 1/2}{(\sqrt{3})^2} = \dfrac{1/2}{3} = \dfrac{1}{6}$ with no errors seen **[A1]**

[Total Marks: 3]

2 **(a)** 2 **[B1]**

(b) $\dfrac{1}{2}$ **[B1]**

(c) $\dfrac{\sqrt{3}}{\sqrt{3}/2}$ **[M1]**

2 **[A1]**

[Total Marks: 4]

3 **(a)** Substitutes $\sin 60° = \dfrac{\sqrt{3}}{2}$ and $\tan 30° = \dfrac{1}{\sqrt{3}}$ (or $\dfrac{\sqrt{3}}{3}$) **[M1]**

1 **[A1]**

(b) Substitutes $\tan 45° = 1$ leading to $\cos x = \dfrac{1}{\sqrt{2}}$ **[M1]**

45 **[A1]**

[Total Marks: 4]

4 **(a)** $\sin 45° = \dfrac{6}{PR}$ (or $\cos 45° = \dfrac{6}{PR}$) **[M1]**

$PR = 6 \div \dfrac{\sqrt{2}}{2} = 6\sqrt{2}$ **[A1]**

(b) $\sin 60° = \dfrac{6\sqrt{2}}{PS}$ **[M1]**

$PS = 6\sqrt{2} \div \dfrac{\sqrt{3}}{2}$ **[M1]**

$PS = \dfrac{12\sqrt{2}}{\sqrt{3}} = \dfrac{12\sqrt{6}}{3} = 4\sqrt{6}$ **[A1]**

[Total Marks: 5]

5 **(a)** $\sin 45° = \dfrac{PQ}{8\sqrt{2}}$ **[M1]**

$PQ = 8\sqrt{2} \times \dfrac{1}{\sqrt{2}} = 8$ **[A1]**

(b) $\cos 30° = \dfrac{8\sqrt{2}}{PS}$ or $PS = \dfrac{8\sqrt{2}}{\cos 30°}$ **[M1]**

$PS = \dfrac{8\sqrt{2}}{\sqrt{3}/2} = \dfrac{16\sqrt{2}}{\sqrt{3}}$ **[M1]**

Rationalises the denominator to get $\dfrac{16\sqrt{6}}{3}$ **[A1]**

[Total Marks: 5]

6 In $\triangle ABD$, $AD = 8\sqrt{2}\cos 45° = 8$ **[M1]**

$BD = 8\sqrt{2}\sin 45° = 8$ **[A1]**

In $\triangle BDC$, $BC = \dfrac{BD}{\sin 30°} = \dfrac{8}{0.5} = 16$ **[M1]**

$DC = \dfrac{BD}{\tan 30°} = 8\sqrt{3}$ **[M1]**

Perimeter $= AB + BC + AD + DC = 8\sqrt{2} + 16 + 8 + 8\sqrt{3}$ **[M1]**

$24 + 8\sqrt{2} + 8\sqrt{3}$ **[A1]**

[Total Marks: 6]

7 (a) Substitutes $\tan 30° = \dfrac{1}{\sqrt{3}}$ to get $\tan^2 30° = \dfrac{1}{3}$ **[B1]**

Evaluates numerator as $1 - \dfrac{1}{3} = \dfrac{2}{3}$ AND denominator as $1 + \dfrac{1}{3} = \dfrac{4}{3}$ **[M1]**

Simplifies $\dfrac{2/3}{4/3} = \dfrac{1}{2}$ which is equal to $\cos 60°$ **[A1]**

(b) Substitutes exact values: $x\left(\dfrac{\sqrt{3}}{2}\right) = \sqrt{3} - x\left(\dfrac{1}{2}\right)$ **[M1]**

Rearranges to collect x terms: $x\left(\dfrac{\sqrt{3}}{2} + \dfrac{1}{2}\right) = \sqrt{3}$ **[M1]**

Isolates x: $x = \dfrac{2\sqrt{3}}{\sqrt{3}+1}$ **[M1]**

Rationalises denominator: $\dfrac{2\sqrt{3}(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)} = \dfrac{6-2\sqrt{3}}{2} = 3 - \sqrt{3}$ **[A1]**

[Total Marks: 7]

8 (a) $\cos 30° = \dfrac{AD}{12}$ **[M1]**

$AD = 12 \times \dfrac{\sqrt{3}}{2} = 6\sqrt{3}$ with no errors seen **[A1]**

(b) Uses Area $= AB \times AD = 60\sqrt{3}$ to find $AB = 10$ **[B1]**

$DE = 12\sin 30° = 6$ **[M1]**

$CE = CD - DE = 10 - 6 = 4$ **[A1]**

(c) Identifies $BC = 6\sqrt{3}$ and $CE = 4$ **[M1]**

$\tan(\angle EBC) = \dfrac{CE}{BC} = \dfrac{4}{6\sqrt{3}}$ **[M1]**

$\dfrac{2}{3\sqrt{3}} = \dfrac{2\sqrt{3}}{9}$ **[A1]**

[Total Marks: 8]

9 (a) Uses cosine rule: $7^2 = x^2 + (x+3)^2 - 2(x)(x+3)\cos 60°$ **[M1]**

Substitutes $\cos 60° = 0.5$: $49 = x^2 + x^2 + 6x + 9 - x(x+3)$ **[M1]**

Expands and simplifies: $49 = 2x^2 + 6x + 9 - x^2 - 3x$ **[M1]**

$x^2 + 3x - 40 = 0$ **[A1]**

(b) $(x+8)(x-5) = 0$ **[M1]**

$x = 5$ (rejects $x = -8$) **[A1]**

Perimeter $= 5 + (5+3) + 7 = 20$ **[A1]**

(c) Uses sine area formula: Area $= \dfrac{1}{2} \times 5 \times 8 \times \sin 60°$ **[M1]**

Substitutes $\sin 60° = \dfrac{\sqrt{3}}{2}$ **[M1]**

$10\sqrt{3}$ **[A1]**

[Total Marks: 10]